

PRANJAL SINGH

Gen AI Full Stack Developer | MERN Stack | RAG | LLM Integration

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PROFESSIONAL SUMMARY

Results-driven **Gen AI Full Stack Developer** and B.Tech CSE student at AKTU with hands-on experience building AI-powered web applications using **React.js, Node.js, Express.js, and MongoDB**. Proficient in **Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, and Vector Databases**. Experienced with **Gemini API, Groq API, JWT Authentication, REST APIs, and cloud deployment** via Vercel and Render. Seeking an internship opportunity to contribute to real-world AI-powered products.

TECHNICAL SKILLS

Frontend	React.js, Next.js, Tailwind CSS, HTML5, CSS3, JavaScript (ES6+), Responsive Design, API Integration
Backend	Node.js, Express.js, REST API Development, JWT Authentication & Authorization, SQL, MongoDB, Mongoose
Gen AI & ML	Large Language Models (LLMs), Prompt Engineering, RAG (Retrieval-Augmented Generation), Vector Databases, Embeddings, Gemini API (Gemini 2.5 Flash, Embedding 001), Groq API, Speech Recognition API
Databases	MongoDB Atlas, MongoDB Atlas Vector Search, SQL, NoSQL Database Design
Tools	Git, GitHub, Postman, Vercel, Render, PDF Processing, Text Extraction

PROJECTS

AI Mock Interview Platform

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, Gemini API, JWT, RAG, Vector Search

- Designed and developed a full-stack **AI-powered mock interview platform** using **React.js, Node.js, Express.js, and MongoDB Atlas** to simulate real-world interview experiences with dynamic question generation.
- Integrated **Gemini API** to dynamically generate **role-specific and skill-based interview questions** from user profile and resume data using **Prompt Engineering** techniques.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** using **Gemini Embedding 001** and **MongoDB Atlas Vector Search** for context-aware, semantically relevant interview Q&A.
- Built secure **user authentication and authorization** using **JWT (JSON Web Tokens)**, enabling interview history tracking and personalized user dashboards.
- Deployed application on **Vercel and Render**, implementing CI/CD workflows for seamless production releases.

College Placement Assistant (AI Chatbot)

Tech Stack: React.js, Node.js, MongoDB Atlas, Gemini 2.5 Flash, RAG, Vector Search, PDF Processing

- Built an **AI-powered placement assistant chatbot** using **React.js, Node.js, and MongoDB Atlas** to help students query placement-related documents using natural language.
- Implemented **PDF upload, parsing, and text extraction** to process company placement materials, eligibility criteria, and preparation guides for ingestion into the knowledge base.
- Architected a **Retrieval-Augmented Generation (RAG) pipeline** leveraging **Gemini Embedding 001** and **MongoDB Atlas Vector Search** for accurate document-grounded question answering.
- Integrated **Gemini 2.5 Flash LLM** to generate contextually relevant responses from uploaded documents.
- Implemented **REST API** endpoints for document ingestion, embedding storage, and chat query handling with **Express.js**.

ChatGPT Clone (Conversational AI App)

Tech Stack: React.js, Node.js, Groq API, Speech Recognition API, REST API

- Developed a **full-stack conversational AI chatbot** with real-time chat capabilities using **Groq API** for ultra-fast **LLM inference** and response generation.
- Implemented **voice input functionality** using the **Web Speech Recognition API**, enabling speech-to-text conversion for hands-free user interaction.
- Designed a **responsive, interactive UI** with persistent chat history, real-time message streaming, and seamless **frontend-backend API integration**.
- Managed **multi-turn conversation context** on the backend to maintain coherent dialogue across sessions.

EDUCATION

B.Tech in Computer Science & Engineering (CSE)	AKTU	Pursuing
Senior Secondary Education (Class XII)	CBSE Board	2023
Secondary Education (Class X)	CBSE Board	2023

CERTIFICATIONS & ACHIEVEMENTS

- **AI Workshop** – COMET, IIT Roorkee (2025) | Gained hands-on exposure to applied AI and machine learning techniques.
- **Java Programming Course** – NPTEL (2025) | Completed certified course in Java programming and object-oriented design.
- **Full Stack MERN + Gen AI Project** – Built production-ready applications combining MERN stack with generative AI capabilities.
- **Cloud Deployment** – Explored and implemented deployment workflows using **Vercel and Render** for production-grade applications.